

965867

Course Name: Digital Image Processing**Course Outcome**

CO1: Understand mathematical formulation of an image, its processing steps and relationship between image pixels.

CO2: Apply Image enhancement using intensity transformations and spatial filtering.

CO3: Analyze image enhancement for frequency domain using Fourier transform.

CO4: Formulate region of interest through morphological operations.

CO5: Evaluate strongly co-related regions obtained through Segmentation using discontinuity and Homogeneity based segmentation techniques

CO6: Describe an object of an image using Shape Number and Boundary descriptors.

Printed Pages:04**University Roll No.****End Term Examination, Even Semester 2023-24****B.Tech (CSE), III year, VI sem (CS ALL)****BCSE0101, Digital Image Processing****Maximum Marks: 50****Time: 3 Hours****Section - A****Attempt All Questions****4 X 5 = 20 Marks**

Attempt All Questions																																																							
No.	Detail of Question	Marks	CO	BL	KL																																																		
1	Consider the two image subsets, S1 and S2, shown in the following figure. For $V=\{1\}$, determine whether these two subsets are a) 4-adjacent, b) 8-adjacent, or c) m-adjacent. S1 <table><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>0</td><td>0</td><td>1</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td></tr></table> S2 <table><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>0</td></tr><tr><td>0</td><td>1</td><td>0</td><td>0</td><td>1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td></tr></table>	0	0	0	0	0	1	0	0	1	0	1	0	0	1	0	0	0	1	1	1	0	0	1	1	1	0	0	1	1	0	0	1	0	0	1	1	1	0	0	0	0	0	0	0	0	0	0	1	1	1	4	1	A	P
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2	Suppose that a 3-bit image ($L=8$) of size 10×10 pixels ($MN = 100$) has the intensity distribution shown in the following table A. Get the histogram transformation function and make the output image with the specified histogram, listed in the table B.	4	2	A	P																																																		

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3	Equalize the given histogram: <table border="1"> <tr> <td>Gray levels (rk)</td> <td>0</td> <td>1</td> <td>2</td> <td>3</td> <td>4</td> <td>5</td> <td>6</td> <td>7</td> </tr> <tr> <td>Number of pixels (nk)</td> <td>50</td> <td>100</td> <td>100</td> <td>300</td> <td>200</td> <td>200</td> <td>50</td> <td>0</td> </tr> </table>									Gray levels (rk)	0	1	2	3	4	5	6	7	Number of pixels (nk)	50	100	100	300	200	200	50	0	4	2	U	P																										
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4	What are the different steps followed by the complete process of edge map generation?									4	5	R	F																																												
5	Suppose an image is given in which user is interested to find the pixels that lie inside the boundary? Devise boundary extraction algorithm and find the boundary pixels. <table border="1"> <tr><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>1</td><td>0</td><td>1</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr> </table> Image <table border="1"> <tr><td>1</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>1</td></tr> <tr><td>1</td><td>1</td><td>1</td></tr> </table> Structure element									1	1	1	0	1	1	1	1	1	1	0	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	4	4	R	C
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Section - B

3 X 5 = 15 Marks

Attempt All Questions

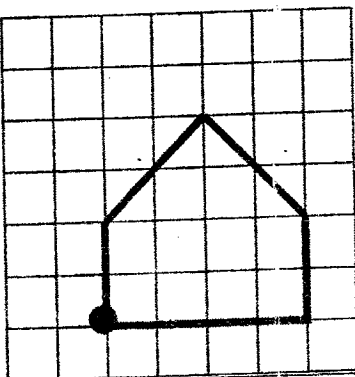
Attempt All Questions																														
No.	Detail of Question	Marks	CO	BL	KL																									
6	What is region splitting and merging method? Explain with suitable example.	3	5	U	F																									
7	What are the different thresholding methods used for segmentation? Explain.	3	5	R	F																									
8	Compute the magnitude and gradient of the shaded pixel. Apply Prewitt operator. <table><tr><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr><tr><td>7</td><td>8</td><td>9</td><td>10</td><td>11</td></tr><tr><td>9</td><td>10</td><td>11</td><td>12</td><td>13</td></tr></table>	1	2	3	4	5	3	4	5	6	7	5	6	7	8	9	7	8	9	10	11	9	10	11	12	13	3	5	A	C
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9	Which operator detects diagonal edges better than Prewitt operator? Also write operator masks.	3	5	R	C																									
10	An image of size 5 x 4 and structuring element of size 1 x 2 are given below. Apply erosion, dilation operation to find the result. <table><tr><td>1</td><td></td><td></td><td></td></tr><tr><td>1</td><td></td><td></td><td></td></tr><tr><td></td><td>1</td><td>1</td><td>1</td></tr><tr><td></td><td>1</td><td></td><td></td></tr><tr><td></td><td>1</td><td></td><td></td></tr></table> Image <table><tr><td>1</td><td>1</td></tr></table> Structuring Element	1				1					1	1	1		1				1			1	1	3	4	A	P			
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Section - C

5 X 3 = 15 Marks

Attempt All Questions

No.	Detail of Question	Marks	CO	BL	KL
11	How the Canny Edge Detection method detects the edge. Explain the complete process.	5	5	R	F
12	Find the following shape-description of the given shape (black circle is the origin and move in the	5	6	A	C

	clockwise direction): i. Chain code ii. First difference iii. Circular first difference iv. Shape number v. Order 																																											
13	There is an image which consists of regions after segmentation. The regions are having gaps which are to be filled. Apply region filling algorithm to fill region's gap. <table border="1" data-bbox="277 844 574 1055"><tr><td></td><td></td><td>1</td><td>1</td><td></td></tr><tr><td></td><td>1</td><td></td><td>1</td><td></td></tr><tr><td></td><td>1</td><td></td><td>1</td><td></td></tr><tr><td></td><td>1</td><td></td><td>1</td><td></td></tr><tr><td></td><td>1</td><td>1</td><td>1</td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr></table> Image <table border="1" data-bbox="316 1106 494 1211"><tr><td>0</td><td>1</td><td>0</td></tr><tr><td>1</td><td>1</td><td>1</td></tr><tr><td>0</td><td>1</td><td>0</td></tr></table> Structuring Element			1	1			1		1			1		1			1		1			1	1	1							0	1	0	1	1	1	0	1	0	5	4	A	P
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